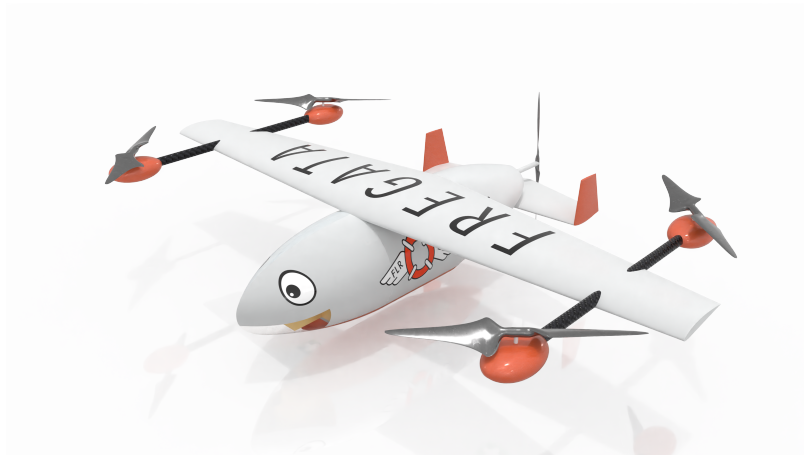




Flying Liferaft

Battery Sizing, Navigational Avionics Selection and Blade Structural Analysis



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Abstract

Fregata is a VTOL UAV with forward flight capabilities aiming to provide humanitarian relief equipment to migrants at sea, specially in the Mediterranean. The payload that it deploys are life-rafts and individual buoys to migrants at sea. The present report presents the battery sizing process and final battery configuration of the UAV. The UAV contains two batteries, one to power the vertical rotors and another one to power the avionics. The battery banks were assembled using cylindrical Lithium-ion cells, and their charging and discharge capabilities were characterised depending on the hover time. Finally, a failure mode analysis for the batteries was carried out, and mitigating processes were applied, such as the addition of a Battery Management System for cell balancing during charging/discharging. The navigational sensors of the system were also determined to fully characterise all of the states of the UAV. These consists of a GNSS coupled with an IMU system, a pitot airspeed sensor, Angle of Attack (AOA) and Angle of Sideslip (AOS) probes, barometric altimeters and LIDAR altimeter sensors. Finally, a structural analysis was carried out on the pusher and vertical propellers subjected to centrifugal forces and bending moments due to aerodynamic forces. In the case of the pusher propeller, the chord distribution was modified to ensure that the material did not fail.

Contents

List of Figures	ii
List of Tables	iii
List of Symbols	iv
1 Introduction	1
2 Battery Sizing	1
2.1 Literature review on current technologies for Hybrid Electric Propulsion	1
2.2 Fuel Cell (FC)	1
2.3 Supercapacitors	2
2.4 Batteries	2
2.5 Battery sizing	3
2.5.1 Power and Energy Requirements	3
2.6 Optimisation process	3
2.6.1 Battery Layout	4
2.6.2 Varying number of cells in series	5
2.6.3 Battery configurations	6
2.6.4 Comparison of sized battery with initial estimate and estimate from energy requirements	6
2.6.5 Discharge Curve Verification and Future Development	6
2.7 Charging	7
2.8 Failure Modes for the Battery	7
2.8.1 Cell balancing	7
2.8.2 Self-discharge	8
3 Systems	8
3.1 Navigational Sensors	8
3.1.1 Geographical Position of the UAV	8
3.1.2 Inertial Measurement Units	9
3.1.3 Sizing of Angle of Attack (AOA) and Angle of Sideslip (AOS) Probe	9
3.1.4 Altitude Sensors	10
3.2 Other additional sensors	10
3.3 Health and Usage Monitoring System (HUMS)	10
3.3.1 Battery Management System (BMS)	10
3.3.2 Engine Sensors	10
3.3.3 Rotor system	11
4 Blade Structural Analysis	11
5 Conclusions	12
References	13
Appendix	15

List of Figures

1	UAV mission profile	1
2	Battery optimisation results	4
3	Battery weight with voltage variation	5
4	DoD and Charge Time with hover/loiter time	7
5	HUMS Configuration	10
6	Stresses due to centrifugal forces	11
7	Stresses due to bending moments	11
8	Ragone Plot [39]	15
9	Energy storage trade-off study	15
10	Main Battery Duty Cycle	16
11	Detailed list of avionics power requirements	17
12	Battery Sizing Optimisation Process	18
13	Battery Discharge Curve for main battery [18]	18

List of Tables

1	Min Battery Power Requirement	3
2	Avionics Power Requirement	3
3	Stacking of cells for both battery banks	5
4	Main Battery Bank Configuration	6
5	Avionics Battery Configuration	6

List of symbols

N_s	Number of cells in series
N_p	Number of cells in parallel
P_{max}	Maximum power
V_{max}	Maximum voltage
V_{cell}	Cell voltage
I_{cell}	Current through a single cell
C_{batt}	Battery capacity
C_{cell}	Cell capacity

1 Introduction

The present report provides a description of the battery sizing methodology and the final battery configuration, as well as a detailed description of the sensors required for navigation of Fregata. Fregata is a fully autonomous VTOL UAV used for maritime search and rescue operations in the frame of migrant ships in the Mediterranean. The aim of the UAV is to provide life-saving equipment, such as life-rafts and individual buoys to maximise survivability until Search and Rescue boats or helicopters arrive.

The mission profile of the UAV can be seen in Figure 1. Note that during the loiter phase, the UAV transitions to hover for payload deployment. The UAV has a parallel hybrid-electric configuration, where batteries are used to power the vertical rotors during VTOL and hover. The UAV has a Maximum Take-Off Weight (MTOW) of 49.5 kg.

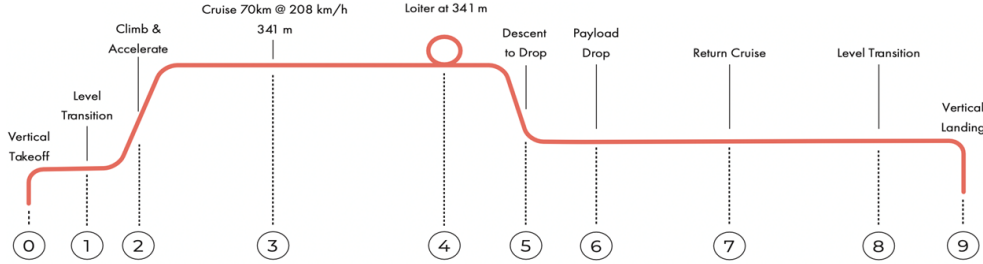


Figure 1: UAV mission profile

2 Battery Sizing

2.1 Literature review on current technologies for Hybrid Electric Propulsion

The first step in the development of the propulsion system for vertical flight involves an in-depth analysis of the current technologies available and their limitations. The main energy source available for electric vehicles (EV) are batteries. However, supercapacitors and fuel cells (FC) can also be implemented. A series of objective criteria were selected for the determination of the optimum energy supply system. It should have a low energy density Wh/kg and a low specific power kW/kg, since in most hybrid-electric UAVs, the batteries represent one of the largest weight penalties [1]. Low specific power is prioritized over low energy density, as power requirements are more constraining than energy requirements as will be shown in Section 2.6.4.

The expected operating life of the product will be 80 missions [2]. With approximately 50 missions per year [2], the UAV is expected to operate 1.5 years. For this reason, technologies with a very large life-span are not prioritised as there is no need to increase the Mean Time To Repair (MTTR) and Mean Time Before Failure (MTBF) of the aircraft above 1.5 years. Additionally, the technology should be relatively cheap and its maturity, measured by the Technology Readiness Level (TRL), should indicate that the technology is fully tested, its performance is fully characterised and there is enough existing supply. This is to ensure that the aircraft can follow the expected design-development-and production timeline [2] which would implement this technology as soon as possible.

2.2 Fuel Cell (FC)

Fuel Cells convert chemical energy stored in fuel to electric energy by the combination of the hydrogen in the cell with oxygen [3]. Currently, the main types of FCs are: Proton-Exchange Membrane FC (PEMFC), Solid Oxide FC (SOFC) and Direct Methanol FC (DMFC) [4].

Experimentally, advantages have been found related to the use of methanol in FCs, mainly high energy density, a reduction in fuel storage weight and low cost. However, its TLR is very low, as it has never been implemented in a UAV powerplant [4]. Similarly, SOFC is a very immature technology [4]. Therefore, only PEMFC are considered in the following trade-off analysis.

PEMFCs have a very high energy density (600-800Wh/kg) and a very low power density (0.5-1 kW/kg [3]). For the take-off manoeuvre, where a power of 11.2kW is required as can be seen in Section 2.5.1, an optimistic weight estimation would require 11.2 kg of FCs, which entails 23% of the MTOW. To obtain this level of power, the cells would need to be stacked in a similar way to batteries, [3]. Due to the high power requirement, this

would occupy a very large volume. There are also high generation costs associated with PEMFCs due to the use of expensive catalysts made of platinum [4], additional weight needed to store the compressed or liquefied fuel and risk of explosion [3]. PEMFCs require the use of qualified maintenance personnel [4]. As a consequence, it is undesirable for the aircraft, as one of its design drivers is a fully autonomous operation from ships with unqualified personnel.

This technology is not fully commercially available for UAV [4] and its performance under non-static and non-standard environmental conditions has not been properly characterised [4]. Since the target implementation timeline is relatively soon, it makes this technology very unsuitable for UAV operations under highly humid and corrosive environments. It has been estimated that the lifespan of FCs will be very high as long as a continuous supply of fuel and oxygen is provided [3].

2.3 Supercapacitors

Supercapacitors are characterised by their high efficiency, their high power density ($\approx 10 \text{ kW/kg}$ [3]) and their large lifespan (greater than 15 years) [3]. This response makes supercapacitors very suited to short, high-power applications such as take-off. This technology would enable a reduction in support and maintenance cost. However, supercapacitors have very low energy densities (10 Wh/kg [5]). From energy requirements set by the Performance, Stability and Control subteam [6], the total energy required is: 342 Wh/kg . This would lead to a supercapacitor mass of 34.2 kg , 69% of the MTOW. The mission stages that require the most energy are the transition and hover stages.

The TLR of supercapacitors is 8-9, because the technology has been fully tested, their performance fully defined and the supply chain fully developed [7]. Currently, there is research being carried out to increase the energy density of supercapacitors, such as those carried out by Superdielectrics. However they are still at the very early stages of research and they have not been put in production [5].

2.4 Batteries

The first design decision carried out regarding batteries, was the use of secondary (rechargeable) batteries. Even though primary batteries can achieve higher power and energy densities, the cost of replacement after each cycle would be too high. The most prominent batteries are: lead-acid, Nickel-Cadmium (Ni-Cd), Nickel Metal Hydride (NiMH), Lithium-ion (Li-ion), Lithium Polymer (LiPo), Lithium Sulphur (LiS) and Lithium-metal [3].

Lithium metal was immediately discarded. They have a high theoretical power and energy density but the development of operable secondary lithium-metal batteries has not been achieved yet [8]. They have very long charging times and are highly prone to failure reducing their life cycle [8]. LiS batteries have a high energy densities (350 Wh/kg [9]). However, its maturity is still not enough for utilisation in the aircraft, since there are strong longevity and performance limitations associated to it [10]. However, it's one of the most promising technologies to substitute Li-ion cells in electric vehicles [10].

Lithium-ion batteries achieve the highest gravimetric energy densities and power densities from the commercially available batteries [11] as can be seen in Figure 8 in the Appendix, which displays the Ragone plot of different technologies. Li-ion cells have a high-capacity use at high currents [11]. This makes Li-ion cells ideal for the main battery, because the battery needs to provide a very large power, meaning that the discharge current will be very high as well. In addition, Li-ion cells do not suffer from memory effects [8] and have a low self-discharge rate [12], meaning that the batteries have better performance across many cycles. Li-ion cells have certain problems related to overcharging/overdischarging and cell imbalance which will be discussed in Section 2.8.

Lithium-ion cells are manufactured into two different types of cells: cylindrical cells and prismatic cells. The most common prismatic cells are pouch-type cells, which are prismatic cells that eliminate the metallic prismatic casing. Both battery banks will be subjected to loads and vibrations resulting from landing impacts and manoeuvre loads. Therefore, the selected battery configuration should be able to withstand certain levels of loads without deforming. Cylindrical cells have the highest resistance to mechanical stress [11]. An additional safety considerations that makes cylindrical cells more suitable is the tendency of pouch cells to swell, which is highly dangerous when operating, as puncturing of the swollen cell can cause it to ignite [13] and damage the rest of avionics. If the battery fails, but no fire occurs, then the battery can be replaced and the UAV will still be able to operate. Cylindrical cells are also more available and have lower associated costs.

The main disadvantages associated to cylindrical cells with respect to pouch cells are the lower packaging density

and the tendency of the centre of the batteries to overheat [11]. The use of cylindrical batteries and the square cell arrangement determined in Section 2.6.1, make the application of cooling mechanisms more efficient.

Therefore, to ensure a safe and stable operations throughout the UAV life and to minimise the risk of in-flight failures, it was decided that cylindrical cells would be used, even if this resulted in a slightly higher weight and volume. A Pugh trade-off study was carried out for the selection of the optimum energy storage technology, as can be seen in Figure 9 in the Appendix. Li-ion cells were the optimum.

2.5 Battery sizing

The drone contains two batteries: the main battery which powers the electric motors and a battery which powers the avionics and other subsystems. The independent configuration was selected due to redundancy and safety concerns. In case of a fault occurring at the main battery, the drone will still be able to operate completely in the forward flight condition, it will still be able to collect data to create diagnostics reports and in the case that an emergency landing is required, it will continue providing light and signal for localisation.

2.5.1 Power and Energy Requirements

The battery is sized to provide continuous power at different stages of flight and its capacity allows completion of one full cycle before recharging is required with appropriate safety margins. It is assumed that at each stage, the power provided by the battery remains constant. The main batteries will only be in operation a total of 6 min: 2 min for VTOL and a maximum of 4 min for hover [2]. The power requirements for the main battery were obtained from the blade power requirements at the different mission stages [14]. These power requirements are summarised in Table 1. The duty cycle of the main battery is shown in Figure 10 in the Appendix.

Table 1: Min Battery Power Requirement

Mission Stage	Take-off	Hover	Landing
Operation Time (min)	1	4	1
Power Consideration (kW)	11.2	8.12	11.2

For the determination of the continuous power requirements for the Avionics Battery, a detailed list of the required power, voltage draw and expected operating time was developed for all avionics. This can be found in Table 11 in the Appendix. The estimation of the avionics battery mass was difficult to accurately predict, since the exact duty cycle of the battery is not known. Initially, it was assumed that all avionics were used during the entire life cycle, meaning the battery needed to provide 167.2 W for 60 min which yielded a battery mass of 1.04 kg. A more accurate battery mass was obtained by considering an estimation of the avionics's operating time. The operating time was classified into three time estimations: 1 h for avionics that will be operating throughout the full mission profile, 10 mins for avionics related to hovering and 1 min for those that have more punctual use. For battery estimation, the worst case scenario of power consumption was assumed. This means that during the 1 min stage all avionics are operating. This is summarised in Table 2. This resulted in a mass reduction of 280 g as shown in Table 5.

Table 2: Avionics Power Requirement

Operating Time (min)	50	10	1
Power Consideration (W)	108.6	158.6	167.2

2.6 Optimisation process

The objective function to be minimised during the optimisation is the weight of the batteries. This is found by determining the optimum number of cells in series and in parallel required to provide a continuous power for a specified time, as described in the previous section. This process is constrained by the maximum voltage of the components the batteries are connected to. For the main battery, this corresponds to the 46.2 V of the electric motors [15] and for the avionics battery, this corresponds to the 28 V of the Satcom [16]. The voltage constraint was selected, as it was determined that the use of a step-up transformer applied to the battery output would lead to a great increase in the weight of the overall electrical system, even though it may be related to a reduction in the battery mass. This assumption was later verified as can be seen in Section 2.6.2

The battery was designed, such that the depth of discharge of an individual cell never fell above 80%. This is to account for the reduction of capacity in high current operations and to ensure that the batteries operate in the linear region of the discharge curve, so no sudden drops in voltage occur. This sudden drop leads to accelerated aging and overdischarge. In the highly non-linear regions, the energy provided by the battery is significantly decreased. A value of 80% was chosen as an initial capacity estimate, as the exact discharge curve varies depending on the individual cell. It was later verified in Section 2.6.5 that for both cells, this maximum depth of discharge was an appropriate estimate by comparing to specifications of similar manufacturers.

The next step involves the determination of the number of cells (N_s) in series obtained from voltage requirements and the number of cells in parallel (N_p) from maximum power (P_{max}) requirements. For the voltage cells, it was assumed that the cells would be supplying power at the nominal voltage. From manufacturer's data, the maximum and standard continuous cell discharge currents are obtained. For the main battery, since the maximum power will be operating for 2 min, this current was chosen for the cell discharge current. For the avionics battery, since it will be operating for longer, the standard continuous discharge current was used instead.

$$N_s = V_{max}/V_{cell}, \quad N_p = \frac{P_{max}}{V_{max}}/I_{cell} \quad (1)$$

Following, this initial configuration, the battery capacity (C_{batt}) is determined from the cell capacity (C_{cell}):

$$C_{batt} = C_{cell} * N_p \quad (2)$$

Then, the capacity discharge related to each operating regime is found by multiplying the discharge current in this section by the discharge time. This was compared to the total battery capacity. If the sum of the capacity discharges was greater than the total battery capacity, an additional cell in parallel was added to the configuration to increase the capacity. The previous process was then iterated until the capacity sum was sufficient. The total weight of the battery pack is then found. This process is iterated for more than 70 cells. Figure 2 figure shows the result for the 15 cells with minimum weight for the main and avionics battery. A flowchart summarising the optimisation process is shown in Figure 12 in the Appendix.

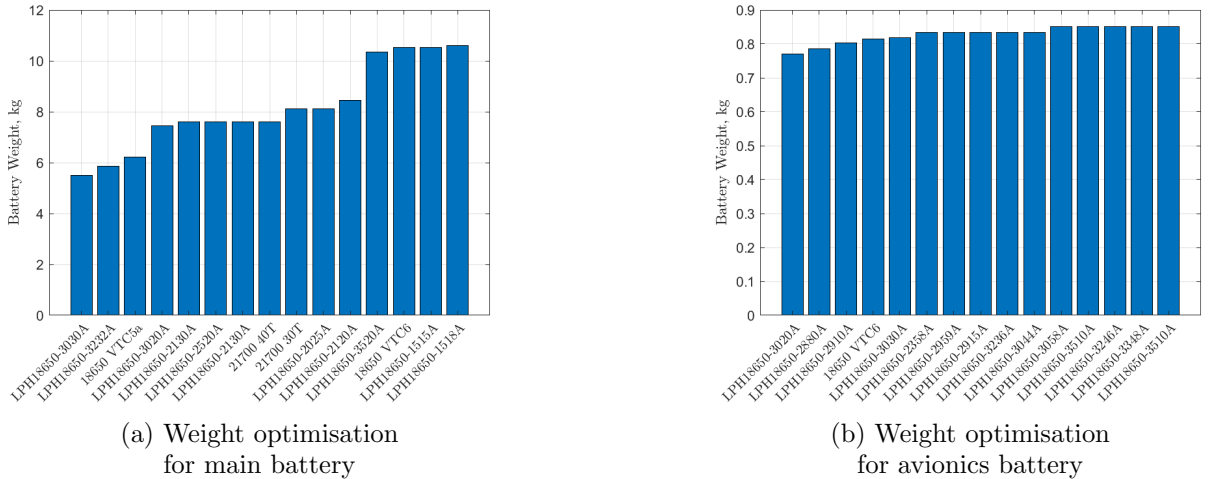


Figure 2: Battery optimisation results

2.6.1 Battery Layout

The cells need to be stacked to form the battery bank. Since for both battery banks the Li-ion cell type is 18650, the cylindrical cell dimensions are 18 mm diameter and 65 mm height. These can be stacked into vertical layers, as can be seen in Table 3. From fuselage layout considerations [16], any length of the batteries should not exceed the fuselage diameter of 0.352 m with an appropriate certain margin (i.e. length of battery 0.3 m). It can be seen that the single layer fits the dimension constraints, minimises volume occupied and reduces complexity of connections. As a result, these dimensions were selected.

It was decided that the batteries will be stacked in a square pattern instead of an hexagonal honeycomb pattern, like the car batteries assembled by Tesla [17]. Honeycomb structures have a higher packaging density. To reduce the introduction of mechanical stresses that could puncture the battery or break the terminal connections, the battery must be strongly fixed to the structure. This can be easily achieved with simple cuboid enclosures. To maintain the volume saved by using the honeycomb structure, a complex geometry casing would need to be used, which would unnecessarily increase manufacturing costs and increase complexity in the mechanism to fix the battery on the fuselage and the mechanism to dampen vibrations. The square arrangement also allows for better heat dissipation between batteries and reduces slightly heat propagation between cells.

In addition, due to the short number of cycles that the UAV will perform, there is no need to separate the battery in modules that can be easily replaced. As a result, it was decided that the increase in packaging density related to a hexagonal arrangement or modular replacement was not advantageous compared to a more simple square packaging pattern.

Table 3: Stacking of cells for both battery banks

Battery Bank	Number of vertical layers	Dimensions	Volume, $\times 10^3 \text{ m}^3$
Main battery	1	$0.237 \times 0.164 \times 0.065$	2.52
	2	$0.128 \times 0.164 \times 0.130$	2.71
	3	$0.091 \times 0.164 \times 0.194$	2.90
Avionics battery	1	$0.146 \times 0.036 \times 0.065$	0.34
	2	$0.073 \times 0.036 \times 0.130$	0.34

2.6.2 Varying number of cells in series

In the previous analysis, the voltage of the battery was constrained to be the same as the maximum voltage of the components connected to it. This was set to avoid the use of a step-up transformer which would have increased the weight of the electrical system. After determining the battery configuration, an analysis was done to ensure that this initial assumption was accurate. The optimisation algorithm was modified to accept an array of possible voltages in steps of 3.6 V (nominal voltage of Li-ion cells). The corresponding configuration for all cells and all voltage configurations was then found.

The weight variation and number of cells in parallel required with battery voltage for the LPH18650-3030A cell is shown in Figure 3 below. From fuselage dimensions constraints, the height, width and length dimensions of the battery should be less than 30 cm. Since cells can be stacked into layers, the maximum number of cells in parallel that was deemed physically possible was 45 cells.

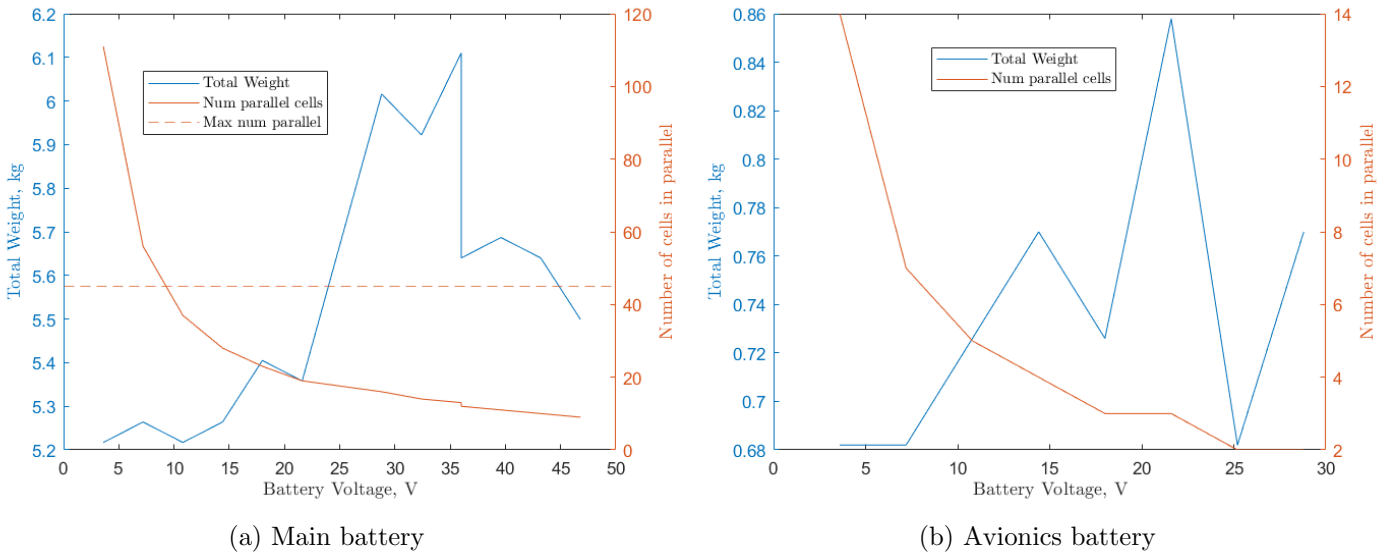


Figure 3: Battery weight with voltage variation

The minimum weight for the main battery is obtained with a voltage of 21.6 V, and that corresponds to a weight

reduction of 141 g with respect to the 46.8 V case. For the avionics battery, the minimum weight is obtained with 25.2 V and that is a 90 g weight reduction. Two step-up transformer would be needed. One that converts 21.6 V to 46.2 V and another 25.2 V to 28.8 V. Step-up transformers operate in alternating current, therefore an additional AC/DC and DC/AC converter would be needed at the input and output of the transformer. The additional weight related to these components overcomes the weight reduction associated with the reduced battery voltage.

2.6.3 Battery configurations

The final configuration for the main and avionics battery are shown below. The battery voltage is greater than the required component voltage: 46.8 V instead of 46.2 V for the main battery and 28.8 V instead of 28 V. Therefore, a voltage divider is added at the output of the batteries. In the more detailed stages of product design, the assembling of this voltage division circuit for each battery would be required

Table 4: Main Battery Bank Configuration

Name of individual cell	LPH18650-3030A	Battery Capacity (Ah)	27	Battery Voltage (V)	46.8
Number of cells in series	13	Total Weight (kg)	5.5	Current Discharge during VTOL (A)	239.3
Number of cells in parallel	9	Discharge temperature range (°C)	-20 to + 60	Current Discharge during hover (A)	173.5
Total number of cells	117	Dimensions of enclosure W x L x H (m)	0.24 x 0.16 x 0.065	Maximum discharge current (A)	270

Table 5: Avionics Battery Configuration

Name of individual cell	LPH18650-3020A	Battery Capacity (Ah)	6	Battery Voltage (V)	28.8
Number of cells in series	8	Total Weight (kg)	0.77	Current Discharge during 50 min (A)	3.8
Number of cells in parallel	2	Discharge temperature range (°C)	-20 to + 60	Current discharge during 10 min (A)	5.5
Total number of cells	16	Dimensions of enclosure W x L x H (m)	0.146 x 0.036 x 0.065	Current discharge during 1 min (A)	5.8
Maximum discharge current (A)	40				

2.6.4 Comparison of sized battery with initial estimate and estimate from energy requirements

In the initial sizing process of the battery, a hover time of 10 min was considered (the whole duration of loiter was hover). By applying the previous analysis to the initial sizing power requirements, the resulting battery mass fraction was 21 %. This corresponds to a battery weight of 10.8 kg. This is twice the current configuration battery weight. This disparity is caused by the significant reduction in hover that resulted from a more detailed definition of the mission profile. This comparison shows that the current batteries are sized to fit precisely the current mission profile. Any increase in hover time required from a re-purposing of the UAV would require a discharge capacity analysis with time to ensure that the maximum depth of discharge is not achieved and a probable resizing of the batteries. From energy requirements, the Performance, Stability and Control subteam calculated that a weight of 1.49 kg was required [6]. This is much lower than the current 5.5 kg. Consequently, can be seen that power requirements were much more constraining.

2.6.5 Discharge Curve Verification and Future Development

Manufacturer's of similar cells have carried out tests to analyse the actual variation of voltage vs capacity under different current conditions [18]. After 2.5 V volts, there is a cut-off voltage, as sudden drops in voltage occur. This leads to a sudden increase in current that creates problems related to accelerated aging and overdischarge. As it can be seen from Figure 13 in the Appendix, the maximum nominal capacity that can be achieved where there is an approximately linear relation is 90 % for the 5 A discharge and 84 % for the 30 A discharge. This is very similar to the initially estimated 80 %. For the avionics, the optimisation code was rerun with this new maximum DoD and the same configuration resulted. This means that the current configuration of the avionics can provide charge for more time than initially designed for.

2.7 Charging

A description of the battery charging characteristics and performance is essential to ensure that the battery is able to fully charge between consecutive uses and to prevent the battery from suffering from accelerated aging and other failures associated with undercharging/overcharging. To charge both battery banks, manufacturer specifications recommend two charging currents: the standard and fast charge currents. Since both main battery and avionics battery were found to be of the LPH18650 battery family, the charging current characteristics for both are the same. The standard charge is 1.5 A and the fast charge is 4 A [19]. The charging method is Constant Current-Constant Voltage (CC/CV).

The charging time will depend on the final DoD which in turn depends on the hovering time of the mission carried out. The following figures show the variation of the standard and fast charge time and the DoD with hovering/loitering time for both battery packs. From manufacturer's specifications, the cells of the main battery are estimated to perform 200 cycles (charge + discharge) at the VTOL discharge current [20]. This exceeds the expected number of operations for the drone. For the cells of the avionics battery, the estimated number of cycles increases to 300 cycles [19]

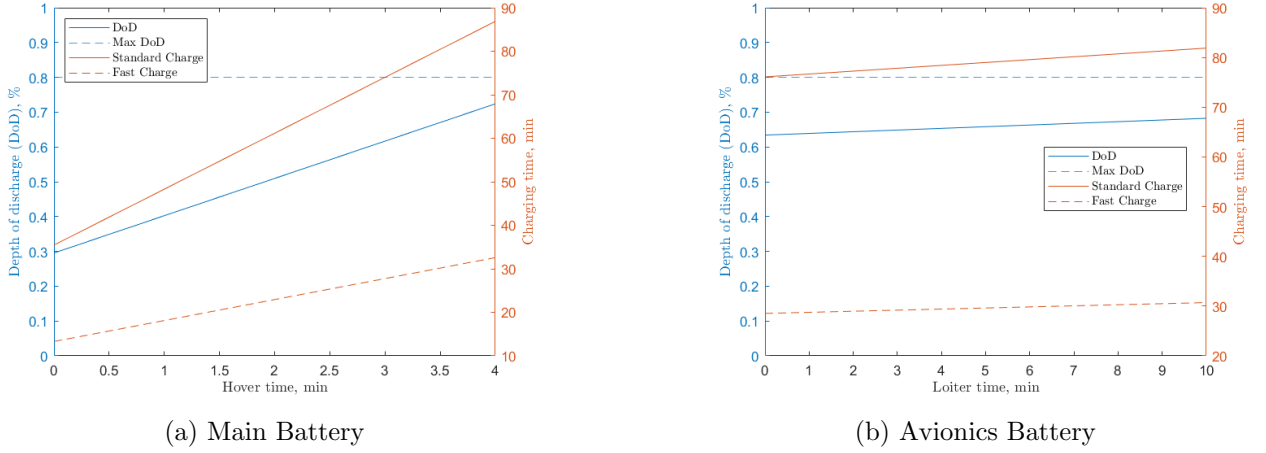


Figure 4: DoD and Charge Time with hover/loiter time

2.8 Failure Modes for the Battery

The battery is expected to function throughout the whole life of the UAV. However, there are a series of failure mechanisms that can prevent this from happenings. The sources of these fault can be external or internal. Mechanical stresses can lead to crushing or puncturing of the cell. Hence, the batteries must be enclosed in a casing that protects the battery by the absorption of the kinetic energy when the structure compresses [21]. A more detailed structural analysis of the loads and vibrations is required to determine the optimum material and geometry of the battery casing. Other external sources of failure are electromagnetic interference resulting from lighting strikes, which are likely due to operation in stormy conditions on the sea. To deal with this the battery is enclosed in a glass fiber reinforced polyester casing as described in the Fregata Structural Layout report [22].

Internal sources of failure include overcharging/undercharging, overdischarging/underdischarging, overheating and internal short circuit [21]. These factors lead to elevated temperature that cause the generation of heat and gas. If the rate of heat dissipation is lower than the rate of heat generation, then thermal runaway will occur [23]. As a result, the battery will heat up and explode. One of the most important methods to prevent this thermal runaway from occurring is to maintain the cells balanced during charging and discharging. An additional characteristic of Li-ion cells that can lead to failure is self-discharge.

2.8.1 Cell balancing

The active materials of the Li-ion cells can be damaged by overcharging/overdischarging of the batteries [24]. Overcharging/overdischarging of the batteries can lead to a large capacity loss and a lower lifespan and thermal stability. One of the main causes of overcharging/overdischarging in battery banks with many cells in series is cell-imbalance. Cell-imbalance can cause one of the cells in series to go beyond the fully charged terminal voltage, even though the battery bank voltage is within acceptable limits. Cell imbalance can be caused by a

disparity in the cell impedance of the cells or a reduction in the capacity of the individual cells due to aging [24]. The effective capacity of the battery is greatly affected by the weakest cell in series. For example, if there is an imbalance of 150mV between cells, this can lead to a reduction in capacity of 25% [25]. As a result, it is imperative to keep the capacity difference between cells in series by 3% [26].

According to manufacturer's specifications the cell imbalance has to be controlled using a voltage sensing system that enables it to control the charging and discharging process and keep the battery's operations within acceptable limits [19] [20]. This can be done by the use of a Battery Monitoring System. The BMS will ensure cell balancing.

2.8.2 Self-discharge

Self-discharge is a characteristic present in Li-ion cells that causes a reduction in the available capacity with time due to unwanted chemical reactions [27]. Exact self-discharge rates are difficult to predict as they are dependent on operating temperature and the state of charge of the battery [28]. Self-discharge rate can be further accelerated by short term exposure to high temperatures. Therefore, during operation, it must be ensured that the cells operate within the bounds stated by the manufacture.

Li-ion batteries have a self-discharge of 5% in the first 24 hrs after discharge and then it losses about 1 - 2 % every month [12]. This means that after the end of the 8th month of storage, the battery capacity will decrease below the minimum required initial capacity to operate the drone (the useful capacity is 80% of the total capacity). Since the UAV is expected to perform 50 missions a year [2], standard self-discharge is not a concern. However, as previously, stated, this can be accelerated by temperature and state of charge conditions and it could become a problem. A careful monitoring of the capacity of the cells must be carried out to ensure an appropriate level of self-discharge is present. This can be achieved by the use of the Battery Management System (BMS).

3 Systems

3.1 Navigational Sensors

Navigational sensors are required to estimate all of the UAV state parameters. The mission is divided into a series of phases each requiring a set of unique sensors: VTOL from moving coastguard ship, navigation in forward flight, and deployment during hovering. In the trade-off analysis carried out, complexity was one of the most important parameters used to select between different types of sensors and between different manufacturer's product. On one hand, the complexity of the component should be enough such that accurate state measurements are done that allow for a safe control of the operations reducing biases and drifts. On the other hand, it is desired that the complexity is low enough so that the power drawn from the battery is reduced, reducing its weight.

The main technologies used nowadays for the position and navigation of UAVs are: stand-alone GPS, augmented GNSS, visual and inertial odometry [29]. During cruising manoeuvres, a GPS/GNSS system will be used to direct the UAV to the target area and to return to base. On the other hand, during hovering manoeuvres, infrared visual odometry will be used for target identification and path planning as described in the Fregata target identification and path planning report [30].

All avionic equipment will be subjected to certain levels of vibrations from the engine and landing, as well as due to manoeuvre loads. Therefore, when selecting COTS components or in future design of avionics, they should be resistant to vibrations or enclosed in a material that dampens the vibrations experienced by the component. In future detail design of the aircraft, these vibrations would need to be quantified.

3.1.1 Geographical Position of the UAV

GNSS sensors allow for highly accurate longitude and latitude positioning of the UAV. The environment along which the drone will be cruising is highly unobstructed, consequently GNSS receivers that require direct line-of-sight can be used [31]. Under these considerations it was determined that a single-frequency GPS system could be utilised, as the need of the system is to locate a generic location (area of shipwreck). For more accurate operations related to the hovering and deployment stage of the mission, the GPS would be coupled with a thermal camera to allow accurate path planning algorithms to be developed. This single frequency stand-alone GPS is the cheapest and most basic configuration.

However, when considering take-off and landing manoeuvres from maritime base stations, a higher level of

accuracy is required [32]. As a result, instead of using a stand-alone GPS configuration an SBAS augmented dual-frequency GNSS system is used. The use of a dual-frequency GNSS receiver minimises error caused by the ionosphere and troposphere. Ionosphere errors are caused when the ultraviolet sun rays ionize gas particles in the atmosphere and they can cause a disrupting effect on electromagnetic GPS satellite signals [32]. Tropospheric errors are delays in the GNSS signal as it moves across this region of the atmosphere. This delay is dependent on local temperature, humidity and pressure [32]. The use of an SBAS augmented system allows an increase in accuracy and provides integrity information. Hovering and dynamic flight testings were carried out by the European Navigation Satellite System (ENGSS), and it was found that by using EGNOS (an SBAS system), the Total System Error reduced by 50% for hover and 80 % for cruising [29]. The tests carried out also demonstrate that there is a reduction in the number of operator interventions when the SBAS system is introduced [29]. Its use in the design fully autonomous drone, will increase the accuracy and safety of its performance. During landing, the GNSS compass moving base mechanism will be used, this requires two GNSS systems

The final chosen GNSS receiver was the ZED-F9T module, a multi-band GNSS receiver with nanosecond-level timing accuracy. The power consumption required is 0.2W [33]. Due to the low power requirements and its negligible weight, the weight contribution of this avionics is minimal. The component has a USB port, which will be used to connect to the microcontroller, as described in the final subsystems layout [16]. The SBAS system allows connection to: WAAS, EGNOS, MSAS and GAGAN. As a consequence, it allows operations in the US, Mexico; Europe and Northern Africa; Japan; and India. This versatility allows for easy launch of the UAV in other geographical locations with minimal adaptations of the avionics.

3.1.2 Inertial Measurement Units

The Inertial Measurement Unit (IMU) is an essential navigational sensor that allows the estimation of the UAV's attitude. For the selection of the IMU model, the following parameters were prioritised: minimising of weight and power draw, high vibration sensitivity, ITAR free, factory calibration, low bias and low bias variation and acceptable heading accuracy (when fitted with a magnetometer). ITAR free refers to the capability of a system to be used without restrictions around the world [34]. This allows easy expansion into other international markets.

The selected IMU is a Micro-Electro-Mechanical Systems (MEMS). This IMU contains accelerometers, gyroscopes and magnetometers. By having an already integrated magnetometer in the system, it allows determination of the yaw angle by comparing the measurement of the magnetic field surrounding the system to Earth's magnetic field. A MEMS IMU was selected, due to their small size, weight and cost, whilst it achieves an acceptable level of accuracy. One of the main disadvantages of MEMS devices (instead of alternatives like Fibre Optic Gyro) is that they have a higher tendency to drift [9]. Since the accelerometers measure linear accelerations and this is integrated once to gain velocity and twice to yield position [9], any accelerometer bias will be lead to a position error which is proportional to the time squared [35]. In a similar way, the gyroscope measures angular rate. This must be integrated to obtain the system's attitude, leading to an error proportional to time if angular rate bias is present [35]. The drift tendency is mitigated by the coupling of the IMU with the GNSS to form an Inertial Navigation System (INS). This combined estimation of the states is enabled by an already incorporated Kalman Filter on the IMU system [9]. There is compatibility in the Input/Output between the IMU and the GNSS.

3.1.3 Sizing of Angle of Attack (AOA) and Angle of Sideslip (AOS) Probe

The main methods used to measure aerodynamic angles in UAVs are the use of a pivoted vane and a differential pressure tube. Pivoted vanes are barely used for UAV navigation due to their high weight [9]. Angle estimation using pressure differential methods involve finding the pressure difference between two orifices oriented at equal angles about the longitudinal axis of the Pitot-Static tube [36]. These angles have to be aligned in the direction of the desired angle. Therefore, for AOS and AOA estimation, a five-hole probe is found to be the most appropriate. Current AOA and AOS pitot tubes are quite expensive due to the need of an air data system to process the information.

At the detailed design stage of the process, avionics engineers would need to design and manufacture this pitot tube and program a micro-controller to process the data. Due to operation in low altitude flight, there is a large risk of icing and water intrusion. The pitot tube will need to be designed to account for this, by including a heating mechanism. The addition of this heating draws quite a lot of power in a short amount of time. A future analysis would need to be carried out to identify whether the current avionics battery is able to provide this amount of power or if a resizing is necessary. An airspeed pitot tube is also used.

3.1.4 Altitude Sensors

A barimetric altimeter is used for altitude estimations during cruising and loiter. To aid VTOL manoeuvres, a LIDAR sensor is used, to ensure that the aircraft is at the required altitude. For the landing mechanism, LIDAR based SLAM is used. The LIDAR, which is oriented downward, will work with the GNSS+INS system to create a map of the landing terrain and determine the most suitable landing position. In the detailed design stages of the production, the algorithm and processing unit required to create this map will need to be constructed.

3.2 Other additional sensors

Other additional avionics include anti-collision green lights and Automatic Dependent Surveillance Broadcast (ADS-B) to track the UAV and provide situational awareness to nearby UAVs.

3.3 Health and Usage Monitoring System (HUMS)

A HUMS system is added to the UAV with the purpose of increasing the reliability, availability and safety of the system. The HUMS consists of a Data Acquisition and Processing Unit (DAPU) that receives data from various sensors. The HUMS then compares it to expected performance and failure modes defined during flight testing and certification phases of the design. The system then creates diagnostics reports that are stored and sends information to the main computer so that it is able to take corrective actions or trigger emergency landing mechanisms. The HUMS increases the MTBF and MTTR of the components analysed. This leads to an overall decrease in downtime and costs, including maintenance and insurance costs.

Due to the necessity to decrease power drawn to the APU (to decrease mass), and the large number of redundancies introduced to the design [16], only the most critical non-redundant components were connected to the HUMS. These were identified to be the engine, rotors and batteries. A diagrammatic configuration of the Health and Usage Monitoring System is shown in Figure 5 below:

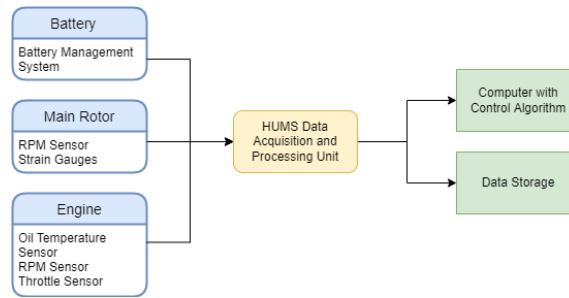


Figure 5: HUMS Configuration

3.3.1 Battery Management System (BMS)

The battery management system is essential for the reliability and long lifespan of Li-ion battery banks. This system performs cell-balancing, voltage and current monitoring, diagnostics and electrical and thermal management [13]. An off-the-shelf BMS, fully compatible with Li-ion batteries with 13s and 8s connections and 28.8 V and 48.8 V was selected. The states that the BMS measures are the State-of-health (SOH), State-Of-Charge (SOC) and State-Of-Power (SOP) [37]. These help with charge/discharge control and cell balancing as described in Section 2.8. This information is fed back to the HUMS that provides the main computer data on how to manage overall energy usage and how to update power capabilities to allow safe and long operation of the batteries [11]. Each terminal in the BMS will be connected to one of the cells in series of the main battery. The BMS will be connected in series with a solid-state relay to allow switching on/off of the batteries when a current from the BMS is provided. The BMS obtains this information from the main computer.

3.3.2 Engine Sensors

The sensors that monitor engine performance detailed in the Engine Control design [15] are also connected to the HUMS. The sensors transmits information to the HUMS such as oil temperature, throttle and RPM setting, and time spent above torque limit.

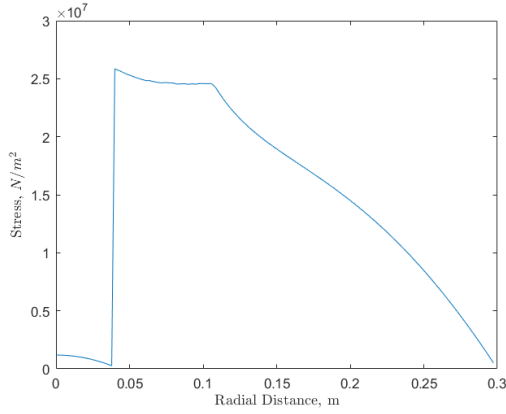
3.3.3 Rotor system

The rotor system is fitted with RPM and strain gauges. These send performance information to the HUMS regarding rotor performance. Strain gauges in the blades allows fatigue monitoring.

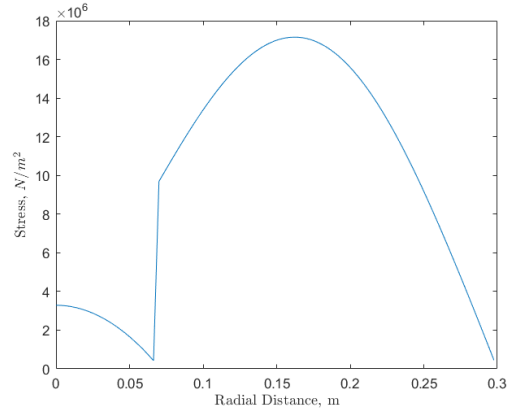
4 Blade Structural Analysis

After the optimal aerodynamic airfoil distribution had been outputted [14], an analytical structural analysis was carried out to determine the blade's ability to sustain stresses due to centrifugal forces and bending moments found using Engineer's Bending Theory for non-symmetric section beams. A material selection analysis was carried out [16], which allowed the material properties such as yield stress and density to be determined.

If the stress at any point in the longitudinal axis of the blade was greater than the material yield stress, the chord from the root to that point was increased by a factor such that the stress at that point decreased over the limit. This process was iterated until the whole blade's stress was below the material yield stress with a safety factor of 2.0. The circular section of the blade was also designed and the start of this region corresponds to the sudden drop in tangential and bending stress at 0.05 m in the radial direction that occur for the pusher and vertical prop. The total stress is the addition of both stresses. Figure 6a and 6b show the stresses due to bending moment for the pusher and vertical props respectively. Figure 7a and 7b show the stresses due to bending moment for the pusher and vertical props respectively.

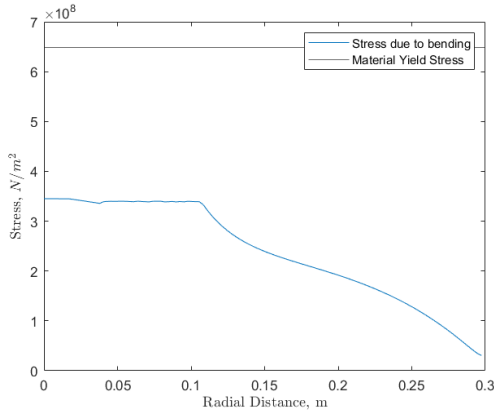


(a) Pusher prop

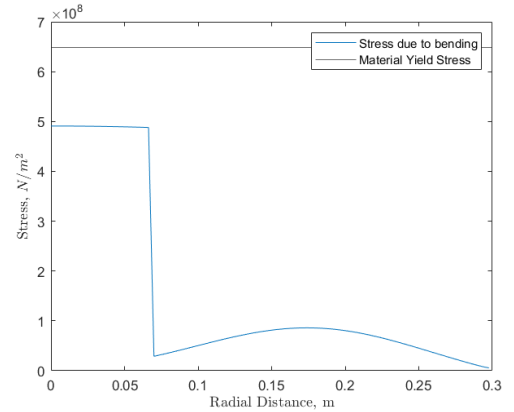


(b) VTOL prop

Figure 6: Stresses due to centrifugal forces



(a) Pusher prop



(b) VTOL prop

Figure 7: Stresses due to bending moments

5 Conclusions

In summary, the batteries for the electric motors and the avionics have been successfully characterised taking into account geometrical constraints, power and capacity requirements. Power requirements were found to be more constraining than energy requirements. To construct the battery bank, cylindrical lithium-ion cells are used. Lithium-ion cells provide the best trade-off between power density, energy density, cost and maturity of the technology. The cylindrical cells are able to withstand greater mechanical stresses, although they increase the mass and weight occupied by the battery. The initial assumption of setting the voltage of the battery bank to be the same or slightly higher than the maximum voltage of the connected components was accurate. This arrangement minimised the overall electrical system weight by avoiding the use of step-up transformers. The capacity loss resulting from high current operations for the main battery was also considered, and it was proven that the initially estimated 80 % capacity loss correlates with experimental tests carried out on similar cells.

The failure modes of the batteries related to external causes such as mechanical stresses and lighting were identified and mitigated by the use of a protective casing. The failures related to charging and discharging are controlled by the addition of the BMS. Even though failures due to normal rates of self-discharge are not probable due to the frequency of use of Fregata, the BMS should still monitor exposure to high temperatures that could lead to accelerated aging and an increase in self-discharge.

In future iterations of the design process, the performance of the battery could be further characterised so that a more accurate estimation of the depth of discharge and power is obtained. At high currents, the discharge curve not only experiences a reduction in capacity as shown in Section 2.6.5, but also experience a reduction in nominal power. Experimental tests would need to be carried out to characterise this behaviour. In reality, the power that the battery needs to produce would be higher than the ones stated in Table 10. This means that more cells in parallel would need to be connected leading to a heavier battery. It would also be necessary to obtain more accurate discharge curves using battery modelling estimation methods. It is specially important to estimate the reduction in capacity and nominal power at different operating temperatures. Another important behaviour that would need to be characterised is the temperature of the cells during the different mission stages to determine whether cooling mechanisms are required. The interface with the charging port and the compatibility with coastguard ship's electrical systems also needs to be further analysed, to ensure that the required charging current can be achieved safely.

Regarding the navigational sensors, low power, low weight and accurate systems were selected to fully define the state parameters of the UAV. GNSS systems were used to guide the UAV to the fixed target location. In order to accurately deploy the lifesaving payload to the migrants, infrared visual odometry is used since it allows for exact environment recognition, path planning and therefore accurate deployment. To increase the accuracy of the inertial measurements, these are coupled with the GNSS. In the more detailed design stages of the product lifecycle, the AOA and AOS pitot tube will need to be designed, which will allow more accurate determinations of power requirements. Once the more detailed avionics power requirements are obtained, these would need to be compared with the current requirements, to determine whether the battery is sufficient or if it needs a slight redesign.

A monitoring system is required to monitor the health of the most critical UAV components: batteries, rotors and the engine. The addition of more sensors is not necessary as these other components have a lower probability of failure and it would increase the weight of the avionics battery.

The stresses due to centrifugal forces are one order of magnitude smaller than those due to bending moments. Therefore, the assumption carried out during the material selection phase [16] of selecting the merit index only based on stresses due to bending moment was accurate. The optimum aerodynamic shape of the pusher propeller blade failed at approximately 0.1 m. Hence, the section from the root until that point was modified. With the modified chord distribution, a Finite Element analysis was carried out [38] to determine more accurate stresses, deflections and torsion characteristics.

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Appendix

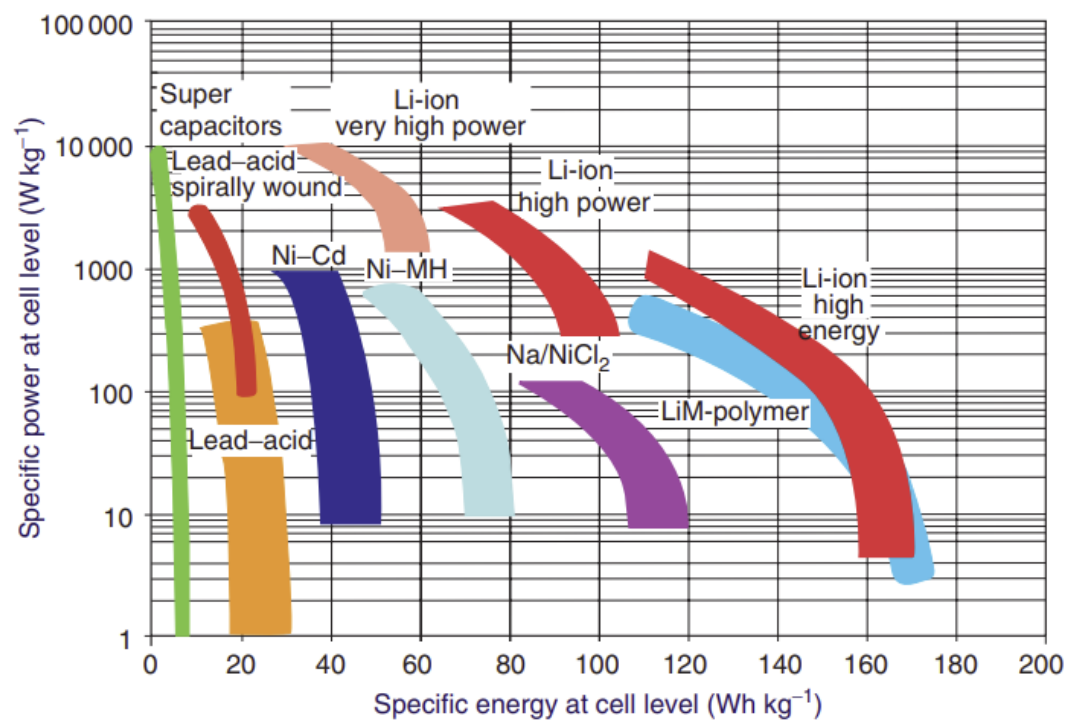


Figure 8: Ragone Plot [39]

Criteria	Importance	Supercapacitors	Fuel Cells (FC)			Battery					
			PEMFC	SOFC	DMFC	Lead-acid	Nickel metal hydride (NiMH)	Nickel Cadmium (NiCd)	Lithium ion (Li-ion)	Lithium Sulfur (LiS)	Lithium Metal
Energy Density (Wh/kg)	8	1	5	5	5	2	2	2	4	5	5
Power Density (kW/kg)	10	5	1	2	1	2	3	2	4	4	4
Cost	7	3	2	1	1	5	5	5	4	1	1
Lifespan	5	5	3	3	3	3	5	3	4	2	2
Technology Readiness Level	9	5	3	2	1	5	5	5	5	2	1
Self-Discharge	6	5	5	5	5	1	2	3	3	3	5
Total Weight		179	136	130	111	137	163	149	183	133	136

Figure 9: Energy storage trade-off study

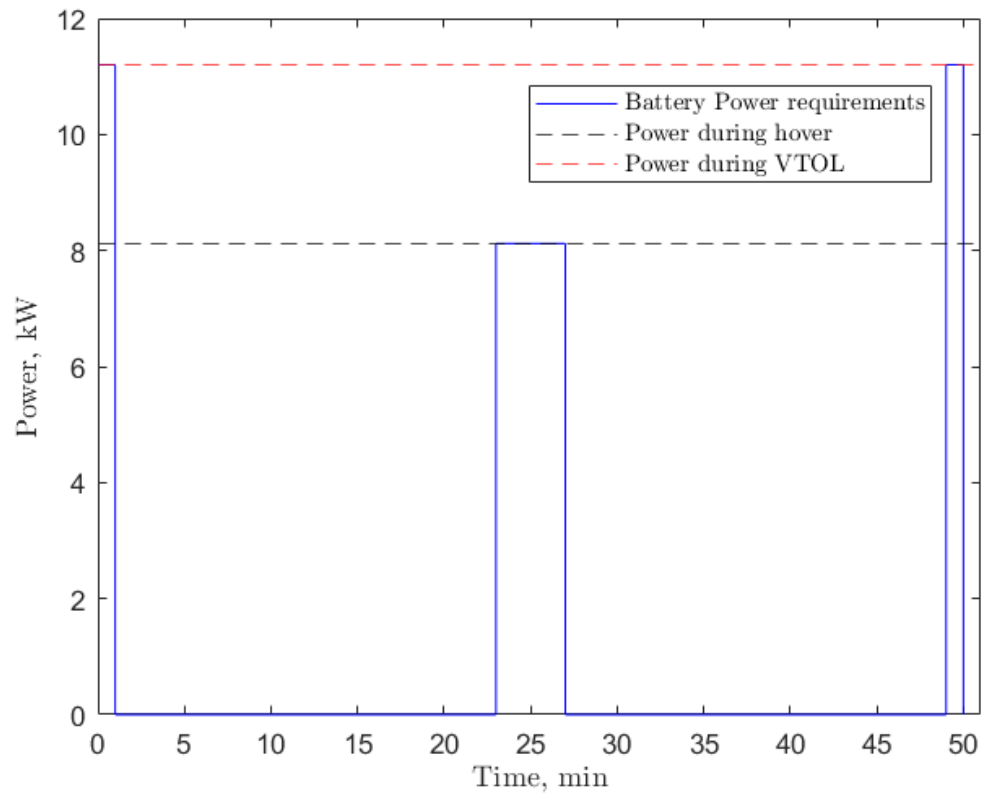


Figure 10: Main Battery Duty Cycle

Components	Quantity	Voltage (V)	Power Required (W)	Time of operation (min)
Communications				
Sat com	1	28	53	60
GNSS Receiver (multi-band)	2	3	0.18	60
Flight Control				
Computer for flight control and processing	2	5	15	60
Microcontroller (python)	6	5	0.25	60
Navigational Sensors				
Barometric Altimeter	2	5	0.1	60
IMU	2	5	0.325	60
Pitot airspeed sensor	2	6	0.15	60
Lidar Altimeter	2	5	1	60
AOA sensor (Pressure differential 4 pitot)	2	5	5	60
Thermal Camera	1	24	8.6	10
Avionics - Actuators				
Aileron servo	2	6	10.9	1
Rudder servo	2	6	2.2	1
Elevator servo	2	6	2.2	1
Payload mechanism servo	2	6	4.3	1
Other Avionics				
EASA green light	2	3.7	0.9	60
RPM Sensor	5	6	0.1	60
Thermal Sensor	2	3.7	0.1	60
Propulsion System				
Clutch	1	24	11	1
Throttle linear actuator	1	12	0.034	1
Total Power for 1h			100.6	
Power with operation time ≤ 1 min			50.056	
Total Power for 10min			8.6	

Figure 11: Detailed list of avionics power requirements

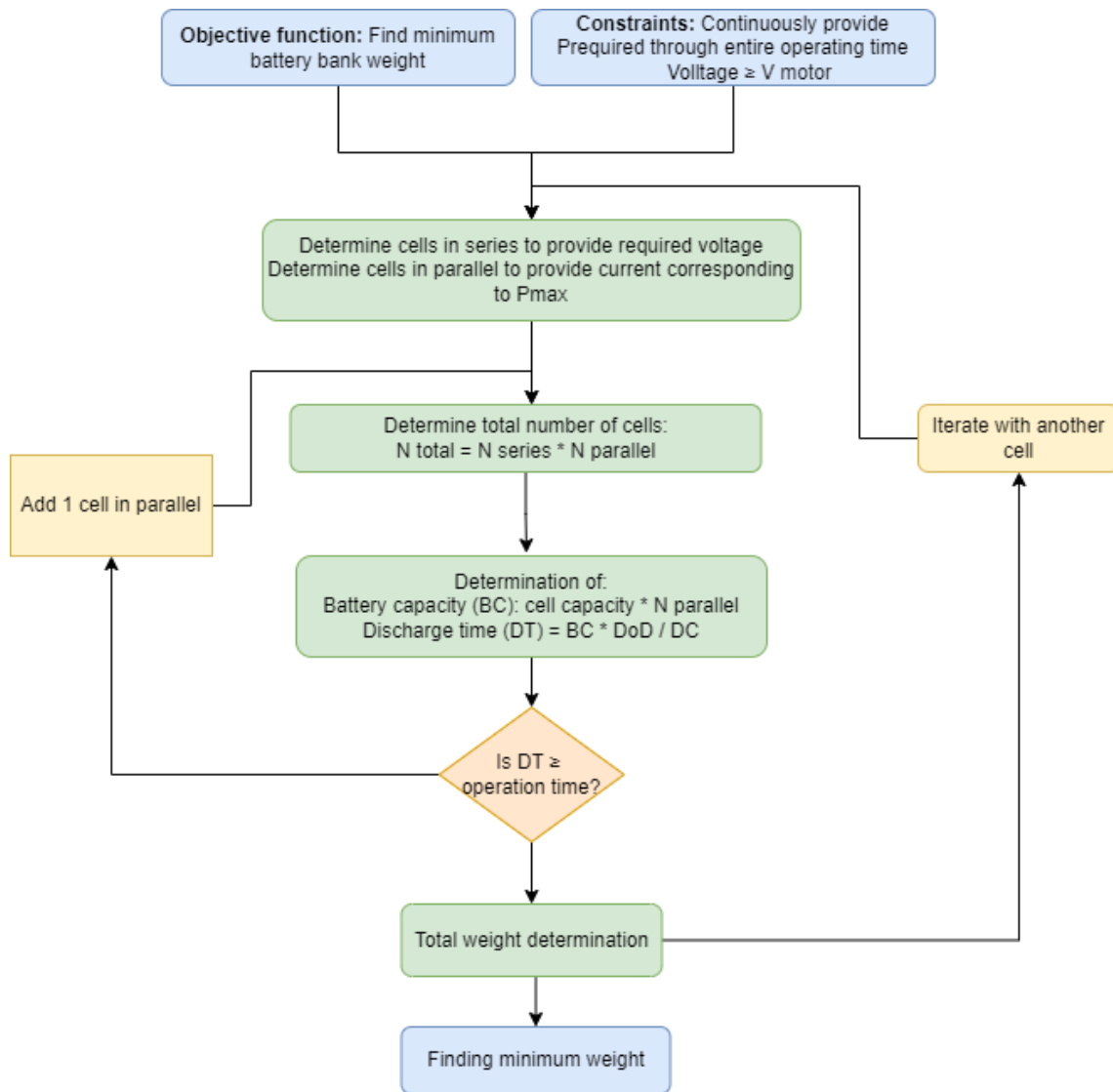


Figure 12: Battery Sizing Optimisation Process

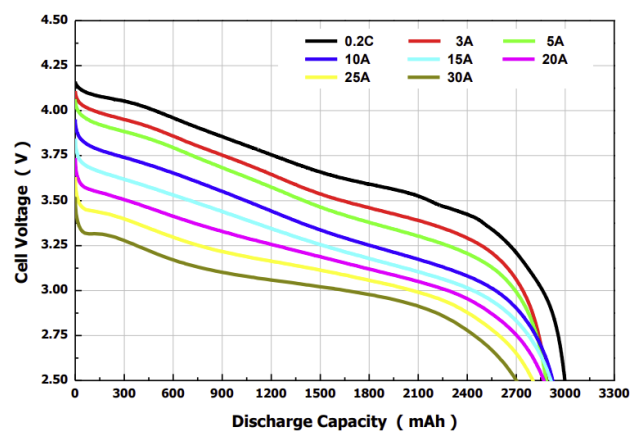


Figure 13: Battery Discharge Curve for main battery [18]